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Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2025

Code: 153603 ***Time: 03 Hours***

Semester: VI ***Subject: Image and Video Processing*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) A gradient image's smallest possible value is:
(i) 0 (iii) -e
(ii) e (iv) 1

(b) What equation do we use for the calculation of the B value according to the HSI components?
(i) B=S(1-I) (iii) B=I(1+S)
(ii) B=I(1-S) (iv) B=S(1+I)

(c) An alternate approach used in place of the median filtering is:
(i) Sharpening (iii) Gaussian filter
(ii) Laplacian filter (iv) Using a mask

(d) Image linear interpolation is given by the formula
(i) v(x,y) = ax+by+cxy+d (iii) v(x,y) = ax+by+d
(ii) v(x,y) = ax+by+cxy (iv) v(x,y) = by+cxy+d

(e) Value of plank's constant h is
(i) $9.55x10^{-34}$ (iii) $7.55x10^{-34}$
(ii) $8.55x10^{-34}$ (iv) $6.55x10^{-34}$

(f) DFT is applied to –
(i) Infinite sequences (iii) Continuous infinite signals
(ii) Finite discrete sequences (iv) Continuous finite sequences

(g) Which of the following fails to work on dark intensity distributions?
(i) Laplacian Transform (iii) Histogram Equalization
(ii) Gaussian Transform (iv) Power-law Transformation

(h) Decoder is used for –
(i) Image Enhancement (iii) Image Decompression
(ii) Image Compression (iv) Image Equalization

(i) In the formula 1-(1/C), C is the
(i) Frame Rate (iii) Constant
(ii) Compression Ratio (iv) Conditional Ration

(j) Which transformation represents shearing in the y-direction?
(i) $(x,y) \rightarrow (x,y)$ (iii) $(x,y) \rightarrow (x+ay,y)$
(ii) $(x,y) \rightarrow (x,y+a)$ (iv) $(x,y) \rightarrow (x,y+bx)$

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Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Explain the steps involved in the digitization of an image. How do sampling and quantization affect image quality?

(b) Define the different types of relationships between pixels. Explain 4-neighborhood, 8-neighborhood, and m-connectivity with examples.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) State and explain any five properties of 2-D DFT useful in image processing

(b) For a 2×2 image f = [[1, 2], [3, 4]], compute the Hadamard Transform.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain Histogram Equalization. How does it enhance image contrast? Derive the transformation function.

(b) Describe smoothing spatial filters. How does an averaging filter work? Provide a 3×3 example.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Compare RGB, CMY, and YIQ color models with respect to representation and applications.

(b) Derive the Inverse Filter in the frequency domain. Why is it highly sensitive to noise?

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Explain the three types of redundancies exploited in image compression: Coding, Spatial, and Temporal redundancy. Provide examples.

(b) Construct a Huffman code for the following symbols and probabilities: A: 0.4, B: 0.2, C: 0.2, D: 0.1, E: 0.1

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Given a 4×1 image: [10, 12, 13, 14], perform predictive coding using predictor: $\hat{x}(n) = x(n-1)$

(b) Explain the differences between Analog Video and Digital Video. Discuss their advantages and limitations.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) A camera captures 640×480 frames at 15 fps, 8 bits/pixel grayscale. Calculate storage needed for a 10-minute video.

(b) Differentiate between Pixel-Based and Region-Based Motion Estimation.

Q.9 Write short notes on any two of the following: **[7 x 2 = 14]**

(a) Discrete cosine Transform

(b) Lossy Vs Lossless Compression modals

(c) Region-Based Motion Estimation

(d) Run Length Coding

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